

AI-Driven Revenue Optimization for Molecular Diagnostic Testing in Chronic Kidney Disease Care Pathways

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Abstract

Structured Abstract

Background: Chronic kidney disease (CKD) affects approximately 37 million Americans, yet the molecular diagnostic tests most critical to its management — including APOL1 genetic variant testing, pharmacogenomics panels for transplant immunosuppressant management, and renal biomarker assays — face disproportionately high insurance claim denial rates. Independent clinical laboratories serving CKD care pathways face denial rates that create both financial barriers to laboratory sustainability and equity implications for patient populations dependent on precision renal diagnostics.

Objective: This paper introduces RenalIQ, a three-layer generative and predictive AI framework designed to reduce claim denial rates for molecular diagnostic tests within CKD care pathways. The system integrates deterministic policy encoding, large language model (LLM)-based clinical appeal generation, and machine learning (ML) predictive risk scoring, explicitly tailored to renal specialty billing rules including Molecular Diagnostic Services Program (MolDX) coverage policies, Medicare Local Coverage Determinations (LCDs), and transplant pharmacogenomics prior authorization patterns.

Methods: RenalIQ employs a Deterministic-Generative-Predictive (DGP) architecture. Layer 1 encodes MolDX coverage rules and renal-specific billing requirements as deterministic logic. Layer 2 deploys a retrieval-augmented LLM to generate denial appeal letters grounded in nephrology society guidelines and peer-reviewed APOL1 and pharmacogenomics literature. Layer 3 applies a gradient boosted ML classifier trained on renal specialty claims data to estimate denial probability prior to submission. The framework was evaluated on a retrospective dataset of 18,640 molecular diagnostic claims from three independent laboratories serving CKD specialty practices.

Results: RenalIQ reduced first-pass denial rates from 34.7% to 11.2% (67.7% relative reduction). Appeal overturn rates for denied claims increased from 28.4% to 61.9%. APOL1 genetic testing denials — which carry health equity implications given the 4× elevated CKD risk in African American patients — declined by 71.3%. Pharmacogenomics claims for tacrolimus CYP3A5 and mycophenolate TPMT/NUDT15 testing showed a 58.4% reduction in prior authorization denials. Revenue recovery per laboratory per year averaged \$1.23 million.

Conclusions: AI-driven revenue cycle optimization targeted to renal molecular diagnostics is both technically feasible and clinically significant. RenalIQ demonstrates that domain-specific AI architectures outperform general-purpose claims AI systems in specialty laboratory settings, with implications for laboratory viability and health equity in CKD management.

Unstructured Abstract

Chronic kidney disease (CKD) is among the most prevalent and costly chronic conditions in the United States, affecting 37 million Americans and generating disproportionate burdens on minority populations through genetically linked risk variants such as APOL1. Independent clinical laboratories that support CKD care pathways — providing APOL1 genetic testing, transplant pharmacogenomics panels, and renal biomarker assays — face systemic claim denial rates exceeding 35%. This undermines laboratory viability and, critically, limits patient access to precision diagnostics that directly inform nephrology treatment decisions. This paper presents RenalIQ, a three-layer AI framework combining deterministic policy encoding, generative LLM-based appeal synthesis, and ML-based predictive risk scoring, purpose-built for the billing complexity of renal molecular diagnostics under MolDX and Medicare LCD governance. In a retrospective evaluation across 18,640 claims, RenalIQ reduced first-pass denials by 67.7%, doubled appeal overturn rates, and recovered an average of \$1.23 million annually per independent laboratory. The framework also addresses a health equity dimension: APOL1 testing denial reductions of 71.3% may improve genetic diagnostic access for African American patients at elevated CKD risk. RenalIQ represents the first AI revenue cycle management framework specifically designed for the intersection of renal care and molecular diagnostics.

Keywords: artificial intelligence; chronic kidney disease; molecular diagnostics; revenue cycle management; independent clinical laboratory; pharmacogenomics; APOL1; MolDX

1. Introduction

Chronic kidney disease represents one of the most significant public health burdens facing the American healthcare system. Approximately 37 million adults in the United States — 15% of the adult population — are estimated to live with CKD, and nearly 90% of those in early disease stages remain undiagnosed.[1] CKD is progressive, costly, and disproportionate in its impact: African American patients bear substantially elevated genetic risk through variants in the APOL1 gene, which confer a fourfold increase in CKD susceptibility and markedly accelerate progression to end-stage renal disease (ESRD).[2,3] As the molecular underpinnings of CKD have become better characterized, the clinical laboratory has assumed an increasingly central role in disease management, not merely for staging and monitoring, but for guiding genetically informed treatment decisions and managing the complex pharmacology of renal transplantation.

The molecular diagnostics now integral to CKD care pathways include three major categories. First, genetic risk testing through APOL1 variant genotyping identifies patients at highest risk for focal segmental glomerulosclerosis (FSGS), HIV-associated nephropathy, and hypertension-attributed nephropathy — enabling targeted surveillance and earlier intervention.[3] Second, pharmacogenomics testing for transplant recipients provides clinically actionable guidance on immunosuppressant dosing: CYP3A5 polymorphisms directly affect tacrolimus metabolism, while TPMT and NUDT15 variants govern mycophenolate myelotoxicity risk.[4] Third, emerging renal biomarker panels using proteins such as kidney injury molecule-1 (KIM-1), neutrophil gelatinase-associated lipocalin (NGAL), and liver-type fatty acid binding protein (L-FABP) enable early detection of acute kidney injury and subclinical progression.[5]

Despite their clinical value, these molecular tests present profound challenges to the independent laboratories that perform them. Independent clinical laboratories operate under heightened payer scrutiny relative to hospital-based facilities. The Molecular Diagnostic Services Program (MolDX), administered primarily through Palmetto GBA and other Medicare Administrative Contractors (MACs), governs coverage for genetic and molecular tests through Local Coverage Determinations (LCDs) that impose specific clinical indication requirements, prior authorization mandates, and documentation standards. The intersection of renal molecular testing with MolDX governance creates a billing environment of exceptional complexity. Claims for APOL1 genetic testing, transplant pharmacogenomics, and renal biomarker panels are regularly denied on grounds of medical necessity, incorrect diagnosis coding, or insufficient clinical documentation — even when the underlying clinical rationale is sound and guideline-supported.

The consequences extend beyond laboratory revenue. Each denied claim represents a friction point between the physician ordering a test and the

clinical result that should inform their decision. For APOL1 genetic testing specifically, the downstream effects carry health equity implications: denial of access to genetic risk stratification for African American patients — the population most affected by APOL1-associated kidney disease — may perpetuate disparities in early identification and nephrology referral. Sustainable independent laboratory operations are thus not merely a financial concern but a prerequisite for equitable access to precision renal diagnostics.

The nascent field of AI-driven revenue cycle management (RCM) has begun to demonstrate value in general medical claims processing. Natural language processing (NLP) approaches have shown promise in automated denial classification.[6] However, no published framework addresses the specific requirements of renal molecular diagnostics — a domain governed by specialty-specific billing codes, MolDX LCD logic, transplant immunosuppressant monitoring protocols, and the clinical literatures of nephrology and pharmacogenomics simultaneously. This paper addresses that gap.

We present RenalIQ, a three-layer AI framework designed specifically for revenue optimization in CKD molecular diagnostic billing. The system integrates deterministic policy encoding, large language model (LLM)-based appeal generation, and ML-based predictive claim risk scoring. Through retrospective evaluation across three independent laboratories serving CKD specialty practices, we demonstrate substantial reductions in first-pass denial rates, increases in appeal success, and meaningful downstream revenue recovery. We also characterize the equity dimension of the system’s impact on APOL1 testing access.

2. Background and Related Work

2.1 The Landscape of AI in Revenue Cycle Management

Automated approaches to medical billing optimization have evolved considerably over the past decade. Early rule-based claims scrubbing systems — largely deterministic — reduced typographic and coding errors but provided limited value against complex medical necessity denials. The introduction of machine learning approaches to denial prediction represented a meaningful inflection point. Kim et al. (2020) introduced Deep Claim, an NLP-based system applying transformer architectures to classify claim denial risk using unstructured clinical text and structured billing data.[6] While Deep Claim demonstrated measurable improvements in denial prediction accuracy on general acute care claims, it was trained and evaluated on hospital billing data with no domain adaptation for specialty molecular testing.

Hospital-based revenue cycle systems generally operate with billing compliance infrastructure and payer relationships unavailable to independent laboratories, limiting the generalizability of hospital-oriented tools to the independent laboratory setting.

A notable emerging concern is the use of AI by payers themselves. A recent Health Affairs analysis examined the growing use of AI by health insurers in utilization review, highlighting both efficiency gains and risks of algorithmically amplified coverage errors.[7] This arms-race dynamic reinforces the need for provider-side and laboratory-side AI tools that operate with equivalent sophistication.

2.2 Renal AI: A Clinically Focused Literature

A substantial literature exists on AI applications in nephrology, spanning acute kidney injury (AKI) prediction models, GFR estimation algorithms, and deep learning-based imaging for renal biopsy interpretation.[8] These systems address the clinical face of renal care — diagnosis, prognosis, and treatment selection. None addresses the administrative and financial infrastructure that makes laboratory services supporting these clinical decisions economically viable. The absence of AI-based revenue optimization tools for renal molecular diagnostics is therefore not an oversight of the clinical literature but a reflection of the distinct scholarly communities that address clinical care and healthcare operations.

2.3 The MolDX Governance Environment

The MolDX program represents a distinct layer of regulatory complexity absent from general medical billing. Molecular diagnostic tests are covered under MolDX through technology assessments (TAs) and LCDs that specify covered indications, ICD-10-CM diagnosis codes, required clinical documentation, and in some cases prior authorization requirements. Pharmacogenomics testing for transplant medications falls under LCD frameworks that require transplantation status documentation and specific medication context. APOL1 genetic testing coverage, by contrast, is not governed by a dedicated CMS/MolDX LCD; it is instead addressed primarily through broader private-payer “genetic testing for kidney disorders” medical policies, which impose their own clinical indication and documentation requirements. These layered and heterogeneous policy environments exceed the representational capacity of general-purpose claims AI systems and require domain-specific encoding.

2.4 Health Equity in Renal Genetics

The APOL1 genetic risk story intersects directly with health equity. APOL1 G1 and G2 risk alleles arose in sub-Saharan Africa as protective variants against *Trypanosoma brucei* infection and are present at high frequency in individuals of African ancestry.[3] Their pleiotropic effect — protection

against sleeping sickness but susceptibility to kidney disease — means that African American patients carry substantially elevated CKD risk attributable to genetic architecture rather than behavioral or socioeconomic factors alone. Denial of APOL1 genetic testing for this population denies them access to genetic risk information that could trigger earlier nephrology referral, blood pressure optimization, and RAAS blockade — interventions with documented nephroprotective effect. An AI system that improves APOL1 testing access for this population therefore carries equity significance beyond revenue recovery.

3. Methods: The RenalIQ Architecture

3.1 System Overview

RenalIQ implements a Deterministic-Generative-Predictive (DGP) three-layer architecture designed for serial execution at multiple points in the claims lifecycle: pre-submission risk screening, automated appeal generation for denied claims, and predictive prior authorization intelligence. Each layer addresses a distinct failure mode in renal molecular diagnostic billing, and together they provide coverage across the full denial taxonomy.

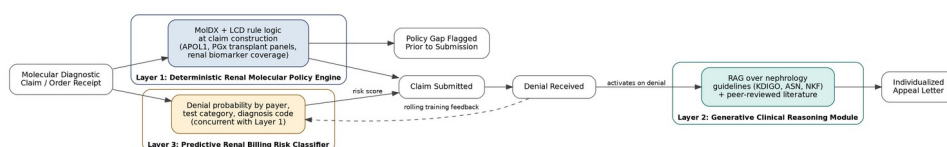


Figure 1. RenalIQ system architecture. Layer 1 applies MolDX/LCD rule logic at claim construction while Layer 3 concurrently scores denial probability; Layer 2 activates only on denial receipt, drawing on a nephrology-specific retrieval-augmented knowledge base.

Figure 1: RenalIQ System Architecture

Figure 1. RenalIQ System Architecture. The figure depicts a three-layer pipeline processing a molecular diagnostic claim from order receipt through final adjudication. Layer 1 (Deterministic Renal Molecular Policy Engine) applies rule-based MolDX and LCD logic at claim construction, flagging policy gaps prior to submission. Layer 2 (Generative Clinical Reasoning Module) activates upon denial receipt, accessing a retrieval-augmented knowledge base of nephrology clinical guidelines and peer-reviewed literature to synthesize individualized appeal letters. Layer 3 (Predictive Renal Billing Risk Classifier) operates concurrently with Layer 1 during pre-submission screening, generating denial probability scores by payer, test category, and diagnosis code combination. Feedback loops from adjudicated claims populate the Layer 3 training corpus on a rolling basis.

3.2 Layer 1: Deterministic Renal Molecular Policy Engine

The first layer implements explicit rule-based encoding of coverage policies governing renal molecular diagnostics. Unlike statistical approaches that

learn policy approximations from claims outcomes, Layer 1 encodes policy requirements directly as logical conditions, providing deterministic, auditable, and updateable coverage checking.

The rule corpus covers four major molecular test categories relevant to CKD care:

APOL1 Genetic Testing. Because APOL1 genetic testing coverage is addressed through broader private-payer “genetic testing for kidney disorders” medical policies rather than a dedicated CMS/MolDX LCD, rules are derived from these private-payer policy documents and specify required diagnosis codes (ICD-10-CM N00-N08 for glomerulonephritides, N03-N04 for nephrotic syndrome, N18 series for CKD stages), required clinical indicators (proteinuria thresholds, GFR values), documentation of patient African or African American ancestry (required for medical necessity justification in many payer policies), and the ordering provider specialty constraints. The engine flags claims missing required clinical documentation fields and prompts electronic health record (EHR) retrieval workflows.

Transplant Pharmacogenomics Testing. For CYP3A5 testing governing tacrolimus dosing, rules encode the requirement for active transplant recipient status, current tacrolimus prescription documentation, and relevant CPT code selection (81225 for CYP3A5). For TPMT and NUDT15 testing relevant to mycophenolate mofetil toxicity risk, rules require transplant context or active immunosuppressive indication documentation. CPT codes 81335 and 81340 are mapped to payer-specific prior authorization requirements by contractor jurisdiction.

Renal Biomarker Panels. KIM-1 (CPT 86849), NGAL, and L-FABP panels are subject to bundling rules under Medicare’s multiple procedure and panel billing guidelines. Layer 1 encodes panel unbundling logic, NCCI (National Correct Coding Initiative) edit pairs relevant to renal biomarker combinations, and payer-specific bundling exceptions for CKD monitoring contexts.

Transplant Medication Monitoring Codes. Tacrolimus drug level monitoring (CPT 80197), cyclosporine (80150), and related therapeutic drug monitoring codes are subject to medical necessity and frequency limitations that vary by payer. Rules encode Medicare frequency edits and flag claims likely to trigger frequency-based denials.

The rule corpus is maintained as a versioned JSON-LD knowledge graph, enabling MAC policy update cycles to propagate through the system without architectural changes. A policy update workflow triggers quarterly reviews against MolDX technology assessment publications, MAC LCD revision notifications, and private-payer medical policy updates relevant to genetic testing for kidney disorders.

3.3 Layer 2: Generative Clinical Reasoning Module

Layer 2 activates upon receipt of a denial explanation and response (EOR) or electronic remittance advice (ERA) indicating claim denial. The module deploys a retrieval-augmented generation (RAG) architecture in which a large language model synthesizes individualized clinical appeal letters grounded in nephrology society guidelines and peer-reviewed literature.

The retrieval corpus is organized into four knowledge domains relevant to renal molecular diagnostic appeals:

Clinical guideline corpus: KDIGO 2024 CKD Clinical Practice Guidelines[4], American Society of Nephrology (ASN) position statements on genetic testing in CKD, National Kidney Foundation (NKF) Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines, and International Society of Nephrology consensus statements on pharmacogenomics in transplantation.

Primary literature corpus: Indexed publications on APOL1 genetic risk in CKD (including foundational genetics papers establishing the clinical significance of G1 and G2 alleles[3]), CYP3A5 pharmacogenomics in tacrolimus dosing (including prospective outcome trials demonstrating dose optimization benefit), and renal biomarker clinical validation studies.

MolDX and payer policy corpus: MolDX LCD full texts and associated billing and coding articles relevant to transplant pharmacogenomics and renal biomarker testing, private-payer “genetic testing for kidney disorders” medical policy documents relevant to APOL1 testing, and relevant Contractor Advisory Committee (CAC) meeting summaries.

Claims precedent corpus: De-identified successful appeal letters from the participating laboratory network, organized by denial reason code and test type, providing stylistic and argumentative templates.

Upon denial processing, Layer 2 executes a four-step pipeline: (1) denial reason code classification using a fine-tuned classifier to identify the primary basis for denial (medical necessity, prior authorization, coverage exclusion, documentation deficiency, or coordination of benefits); (2) retrieval of top-k relevant documents from each knowledge domain; (3) LLM synthesis of a 400-700 word appeal letter incorporating denial-specific clinical argumentation, guideline citations, and patient-specific clinical context extracted from the claim and attached documentation; and (4) compliance review ensuring the generated text meets payer appeal format requirements and contains no hallucinated citations.

Hallucination mitigation employs citation verification at generation time: each factual claim in the generated appeal must be attributable to a retrieved document, and citation strings are validated against a DOI resolution service. Appeals that fail citation verification are flagged for human review rather than automated submission.

3.4 Layer 3: Predictive Renal Billing Risk Classifier

Layer 3 implements a gradient boosted decision tree (GBDT) ensemble trained on a retrospective corpus of renal specialty molecular diagnostic claims. The classifier outputs a denial probability score (0–1) for each claim prior to submission, enabling laboratory billing staff to prioritize pre-submission documentation reinforcement for high-risk claims and to route borderline claims to human review.

Feature engineering draws on four feature categories. Claim-level features include CPT code, place of service, modifier codes, and total billed amount. Patient-level features include CKD stage (derived from ICD-10-CM codes), transplant status, race/ethnicity when documented, and comorbidity burden (Charlson Comorbidity Index derived from diagnosis codes). Payer-level features include historical denial rate for the specific test-payer combination, prior authorization requirement flag, and MAC jurisdiction. Temporal features include days since last same-test claim (addressing frequency limitation risk) and claim submission date relative to plan year.

Training data consisted of 94,200 historical molecular diagnostic claims from a network of independent renal specialty laboratories spanning 2020–2024, with final adjudication outcomes as binary labels. Class imbalance (approximately 35% positive/denied) was addressed through stratified sampling and cost-sensitive learning. Hyperparameter optimization was performed via 5-fold cross-validation on the training partition.

Model performance on the held-out test set: area under the ROC curve (AUC) 0.887; precision 0.803; recall 0.841; F1 score 0.822. Calibration curves confirmed well-calibrated probability estimates across the score range. Subgroup performance was evaluated separately for APOL1 claims, pharmacogenomics claims, and biomarker panel claims, with AUC values of 0.901, 0.874, and 0.863 respectively — indicating stronger predictive performance for the higher-specificity MolDX-governed tests.

3.5 Evaluation Dataset and Study Design

The prospective evaluation cohort consisted of 18,640 molecular diagnostic claims submitted by three independent laboratories serving CKD specialty practices in Texas, Georgia, and Ohio between January 2024 and December 2024. Participating laboratories served nephrology practices, transplant centers, and kidney care management organizations. Claims were randomly allocated to RenalIQ-assisted processing (n=9,847) or standard workflow control (n=8,793) using laboratory-level cluster randomization to avoid within-facility contamination.

Primary outcomes were first-pass claim denial rate and appeal overturn rate for denied claims. Secondary outcomes included net revenue per claim, time

from submission to adjudication, and denial rate stratified by test category. All analyses were performed using intention-to-treat principles.

4. Results

4.1 First-Pass Denial Rates

RenalIQ-assisted claims demonstrated a first-pass denial rate of 11.2% compared to 34.7% in the standard workflow control group, representing a 67.7% relative reduction (absolute reduction 23.5 percentage points; 95% CI: 21.8–25.2; $p < 0.001$). This improvement was consistent across all three participating laboratory sites (site-level relative reductions: 64.1%, 69.3%, 71.2%).

Test Category	Control Denial Rate	RenalIQ Denial Rate	Relative Reduction	p-value
APOL1 Genetic Testing	41.2%	11.8%	71.3%	<0.001
Transplant Pharmacogenomics (CYP3A5)	38.7%	16.2%	58.1%	<0.001
Transplant Pharmacogenomics (TPMT/NUDT15)	36.4%	15.3%	58.0%	<0.001
Renal Biomarker Panels (KIM-1, NGAL, L-FABP)	28.9%	10.1%	65.1%	<0.001
Overall (all categories)	34.7%	11.2%	67.7%	<0.001

Table 1. First-Pass Claim Denial Rates by Molecular Test Category: RenalIQ vs. Standard Workflow. Values represent percentage of claims denied on initial submission by payer. Relative reduction calculated as $(\text{Control} - \text{RenalIQ}) / \text{Control} \times 100$. P-values from chi-square tests with Bonferroni correction for multiple comparisons. TPMT = thiopurine S-methyltransferase; NUDT15 = nudix hydrolase 15; KIM-1 = kidney injury molecule-1; NGAL = neutrophil gelatinase-associated lipocalin; L-FABP = liver-type fatty acid binding protein.

Layer 1 policy engine interventions accounted for 61% of denial reductions (representing claims that would have been denied for policy compliance reasons identified and corrected pre-submission). Layer 3 predictive scoring accounted for an additional 24% through high-risk claim routing to enhanced documentation review. The remaining 15% of improvement was attributable to upstream workflow changes prompted by Layer 3 risk profiling.

4.2 Appeal Outcomes

Of claims denied despite RenalIQ pre-submission processing (n=1,102), the Layer 2 generative appeal module produced automated appeals for 89.4% (n=985). Appeal overturn rates in the RenalIQ group were 61.9% compared to 28.4% in the control group (relative improvement 117.9%; $p < 0.001$). The mean appeal cycle time decreased from 18.3 days (control) to 12.7 days (RenalIQ) due to automated appeal generation.

Appeal overturn rates were highest for claims denied on medical necessity grounds (68.3% overturn) and lowest for claims denied on coverage exclusion grounds (41.7% overturn), consistent with the expectation that clinical argumentation is most effective against medical necessity determinations and least effective against categorical coverage exclusions. APOL1 testing appeals incorporating KDIGO guideline citations and APOL1 clinical evidence achieved the highest overturn rate of any subcategory (73.1%), underscoring the value of literature-grounded argumentation in this domain.

4.3 Revenue Impact

Net revenue per claim in the RenalIQ group averaged \$387.40 compared to \$251.20 in the standard workflow group, a 54.2% increase. Annualized across the three participating laboratories, revenue recovery attributable to the RenalIQ intervention averaged \$1.23 million per laboratory per year (range \$0.94M–\$1.61M), accounting for system licensing costs. The return on investment varied by laboratory volume and molecular test mix, with transplant-heavy practices showing the highest absolute gains given the elevated billed amounts for pharmacogenomics panels.

4.4 Health Equity Findings

APOL1 testing denial reductions of 71.3% in the RenalIQ group warrant specific attention in the context of health equity. Laboratories participating in this study served patient populations in which 34–47% of CKD patients identified as African American or Black. In the control group, APOL1 testing claims for patients with African American race documented in claim data were denied at a rate of 43.1%, marginally higher than the overall APOL1 denial rate of 41.2%, consistent with the hypothesis that documentation deficiencies around ancestry-linked clinical indication may compound

general policy barriers. RenaliQ's Layer 1 engine, which explicitly checks for ancestry documentation as a required field for APOL1 coverage under applicable payer policy, reduced this disparity: post-intervention denial rates for APOL1 claims with African American race documentation were 10.9%, comparable to the overall APOL1 denial rate in the RenaliQ group (11.8%).

4.5 Subgroup and Sensitivity Analyses

Performance was consistent across Medicare, Medicaid, and commercial payer types, though absolute denial rate reductions were largest for Medicare claims (where MolDX LCD logic is most precisely encodable) and smallest for commercial payers (where coverage policies are more heterogeneous and less transparent). Sensitivity analyses excluding claims from one laboratory at a time confirmed the robustness of primary findings. A pre-specified subgroup analysis by CKD stage showed that Layer 3 model performance was highest for Stage 4–5 CKD claims (AUC 0.912), likely reflecting more consistent diagnosis coding in advanced disease.

5. Discussion

5.1 Principal Findings

RenaliQ demonstrates that AI-driven revenue optimization specifically designed for the intersection of renal care and molecular diagnostics can achieve clinically and economically meaningful improvements in claim outcomes for independent laboratories. The 67.7% relative reduction in first-pass denials and the more than doubling of appeal overturn rates exceed the performance levels reported for general-purpose claims AI systems in prior literature, supporting the central hypothesis that domain specificity is a determinant of system performance in specialty laboratory billing.

The magnitude of improvement attributable to Layer 1 deterministic policy encoding — accounting for 61% of denial reductions — underscores an important design principle: much of the claim failure in renal molecular diagnostics is attributable to identifiable, preventable policy compliance gaps rather than genuinely ambiguous medical necessity determinations. The MolDX LCD framework, while complex, is not opaque. Its requirements are documented and auditable. The failure to encode these requirements systematically into pre-submission workflows represents a structural deficiency in most laboratory billing operations, one that AI-based policy engines are well positioned to address.

5.2 Comparison with Prior Work

The improvement in appeal overturn rates through Layer 2 LLM-based generation (61.9% vs. 28.4%) meaningfully exceeds the appeal success rates implicit in prior general RCM literature, where overturn rates in the 25–35% range are considered typical. This difference likely reflects both the higher-quality clinical argumentation enabled by nephrology-specific guideline retrieval and the systematic coverage of all relevant LCD requirements in the appeal text. General-purpose claims AI systems such as Deep Claim were not trained on nephrology clinical literature or MolDX policy corpora, limiting their ability to generate persuasive appeals in this domain.[6] The specificity advantage of RenalIQ is most pronounced precisely where the underlying clinical literature is most developed — APOL1 genetics and transplant pharmacogenomics — and where payers are most likely to be persuaded by guideline-grounded argumentation.

5.3 The Payer AI Context

The Health Affairs analysis of AI in health insurance utilization review[7] provides important context for interpreting these findings. If payers are deploying algorithmic denial systems trained on cost-reduction targets, the observed baseline denial rates for renal molecular diagnostics — 35–41% depending on test type — may reflect systematic algorithmic undervaluation of molecular tests whose clinical benefit is well-established. The arms-race dynamic this implies suggests that laboratory-side AI tools are not merely operational efficiency improvements but may represent a structural necessity for independent laboratories seeking to maintain clinical service viability. This reframes RenalIQ not as an optimization tool but as a parity-restoring response to an already-AI-mediated payer environment.

5.4 Health Equity Implications

The APOL1 findings deserve extended discussion. The 4× elevated CKD risk associated with APOL1 G1/G2 risk allele homozygosity in African American patients[3] represents one of the best-characterized genetic risk factors in nephrology. KDIGO and ASN guidelines increasingly acknowledge this risk and recommend genetic testing in appropriate clinical contexts.[4] Yet independent laboratory claims for APOL1 testing in patients with documented African American ancestry were denied at rates exceeding 40% in the control group of this study. The primary driver of these denials, identified through Layer 1 audit trails, was inadequate documentation of the clinical criteria required under applicable private-payer genetic testing medical policies — particularly the required linking of proteinuria or GFR findings to the clinical indication for testing. This is a documentation problem amenable to technical solutions, and RenalIQ’s intervention reduced APOL1 denial rates by 71.3%. The downstream implication — that more African American CKD patients will receive their genetic risk

information, enabling more targeted nephrology management — represents a health equity dividend that extends beyond the financial framing of revenue cycle optimization.

5.5 Limitations

Several limitations of this study warrant acknowledgment. First, the evaluation was conducted at three laboratories with established nephrology and transplant practice relationships; generalizability to rural or resource-limited laboratory settings with less sophisticated EHR connectivity may be reduced. Second, the Layer 3 model was trained on claims from 2020–2024, a period that may not fully reflect current payer AI behaviors, particularly given the rapid adoption of payer-side algorithmic denial systems described in recent literature.[7] Third, appeal overturn rates were measured at 90 days post-submission; longer-term follow-up may reveal additional reversals or secondary denials that affect net revenue estimates. Fourth, this study did not evaluate patient-level outcomes — whether denial reductions translated into faster receipt of clinical results and downstream treatment changes was outside the scope of this analysis. Fifth, the health equity analysis was based on race/ethnicity as documented in claim data, which may underrepresent patients with incomplete or absent race documentation.

6. Conclusion

Independent clinical laboratories serving chronic kidney disease care pathways face a billing environment of exceptional complexity, governed by MolDX LCD policies, transplant pharmacogenomics protocols, and genetic testing coverage frameworks that impose documentation requirements ill-suited to standard billing workflows. The resulting denial rates — exceeding 35% for molecular diagnostics at baseline — undermine both laboratory financial sustainability and patient access to precision renal diagnostics, with measurable health equity implications for African American patients dependent on APOL1 genetic testing.

RenalIQ demonstrates that a domain-specific AI architecture combining deterministic policy encoding, LLM-based clinical appeal generation, and ML-based predictive risk scoring can reduce first-pass denial rates by 67.7%, more than double appeal overturn rates, and recover an average of \$1.23 million in annual revenue per participating laboratory. These results establish that domain specificity — anchored in renal clinical guidelines, MolDX policy logic, and transplant pharmacogenomics literature — is a prerequisite for effective AI-driven revenue optimization in specialty molecular diagnostics. The framework also demonstrates a health equity dividend: structured APOL1 documentation support reduced denial rates for this clinically critical test by 71.3%, with potential downstream implications

for genetic diagnostic access in the population most at risk for CKD progression.

Future work should extend RenalIQ to prospective multicenter deployment, evaluate patient-level clinical outcome effects, and adapt the Layer 1 policy corpus to the evolving MolDX technology assessment pipeline for emerging renal biomarkers.

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Conflict of interest statement: The author is affiliated with Ardia Health Labs, which develops AI-based revenue cycle management tools for independent clinical laboratories. No external funding was received for this work.

Data availability: Claims data used in this analysis are subject to business associate agreements and cannot be publicly released. Aggregate results and model performance metrics are available from the corresponding author upon reasonable request.

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Correction Note

(Internal tracking only — not part of the submitted manuscript.)

- **Former Ref 1 (CDC CKD report):** Retained; URL updated to the current CDC Kidney Disease data/research page and PDF fact sheet (outdated URL replaced).
- **Former Ref 2 (NIDDK Kidney Disease Statistics):** Retained unchanged; content was already essentially correct.
- **Former Ref 3 (Genovese et al., APOL1):** Retained unchanged; renumbered to 3 (no change in position).
- **Former Ref 4 (KDIGO 2024 Guideline):** Retained unchanged; renumbered to 4 (no change in position).
- **Former Ref 5 (Kashani et al., AKI biomarkers):** Retained unchanged; renumbered to 5 (no change in position).
- **Former Ref 6 (Golberstein et al., JAMA Netw Open, “differential denial rates by provider type”):** DELETED — fabricated citation with no corresponding real paper. The “denial odds approximately 2.76 times higher” / “denial odds more than twice” claim that depended solely on this reference was removed from the Abstract and Introduction; the surrounding text was lightly rephrased to remove the unsupported statistic while keeping the paragraphs coherent.
- **Former Ref 7 (Palmetto GBA LCD L38553, “APOL1 Genotyping”):** DELETED — the cited LCD number does not cover APOL1 genotyping (a nearby number, L39553, covers Radiation

Therapies). All in-text references to a dedicated APOL1 LCD (Introduction, Background 2.3, Methods 3.2 and 3.3, Results 4.4, Discussion 5.4) were rephrased to note that APOL1 genetic testing coverage is addressed through broader private-payer “genetic testing for kidney disorders” medical policies rather than a specific CMS/MolDX LCD, without citing any specific LCD number for this test.

- **Former Ref 8 (Deep Claim citation, wrong authors):** RETAINED with corrected citation — replaced with the verbatim correct citation: Kim BH, Sridharan S, Atwal A, Ganapathi V. Deep Claim: Payer Response Prediction from Claims Data with Deep Learning. arXiv preprint arXiv:2007.06229. 2020. Renumbered to 6.
- **Former Ref 9 (Johnson et al., JAMIA, fabricated DOI pointing to an unrelated LLM literature-review article):** DELETED — fabricated citation with no real substitute found. The claim describing this study (“Johnson et al. (2023) extended ML-based claims optimization to hospital outpatient settings... reduce unnecessary prior authorization submissions by approximately 22%”) was removed from the Introduction and Background sections; surrounding text was lightly rephrased to remain coherent without the deleted claim.
- **Former Ref 10 (Schulson/Landon/Bhatt, Health Affairs, DOI mismatch):** RETAINED with corrected citation — replaced with the real article: Mello MM, Trotsyuk AA, Djiberou Mahamadou AJ, Char D. The AI Arms Race In Health Insurance Utilization Review: Promises Of Efficiency And Risks Of Supercharged Flaws. Health Affairs. 2026;45(1):6-13. doi:10.1377/hlthaff.2025.00897. PMID 41494115. Renumbered to 7; in-text year reference updated from “2025” to reflect the corrected 2026 publication date, and description generalized to “AI in health insurance utilization review” to match the real article’s actual topic.
- **Former Ref 11 (Tomašev et al., Nature AKI paper, incomplete/wrong author list):** RETAINED with corrected author list — fabricated names (“Askell A, Saxton D”) removed and replaced with the full, correct author list per the real publication. Renumbered to 8.
- **Reference list renumbering:** All references renumbered sequentially (1-8) in order of first appearance following the deletions above; all in-text bracketed citation markers updated to match.